

# SANGWON JEONG

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## SUMMARY

PhD candidate in Computer Science specializing in explainable AI, generative models, LLM evaluation/forecasting, and visual analytics. Builds methods and systems for analyzing, evaluating, and steering generative model behavior, with research spanning concept-based interpretability, representation analysis, visualization systems, and structured-output generation. Published in TVCG, IEEE VIS, and EuroVis, with experience developing research prototypes, scalable evaluation pipelines, and interactive tools for AI model understanding. Current work on forecasting LLM outputs pre-generation is under review at ICLR 2027.

## EDUCATION

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|--------------------------------------------------------------|---------------|
| <b>PhD in Computer Science</b><br>Vanderbilt University      | December 2026 |
| <b>MSc in Computer Science</b><br>Vanderbilt University      | May 2020      |
| <b>BA in Business Administration</b><br>Seokyeong University | March 2018    |

## UNDER REVIEW

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|----------------------------------------------------------------------------------------------------------------------------|------|
| <b>Bayesian Forecasting of LLM Outputs from Pre-Generation Hidden States</b><br>S. Jeong et al., Under review at ICLR 2027 | 2026 |
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## PUBLICATIONS (FIRST AUTHOR)

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| <b>Concept Lens: Visual Comparison and Evaluation of Generative Model Manipulations</b><br>S. Jeong et al., IEEE Transactions on Visualization and Computer Graphics (TVCG)            | 2025 |
| <b>Text-based Transfer Function Design for Semantic Volume Rendering</b><br>S. Jeong et al., IEEE VIS                                                                                  | 2024 |
| <b>CAN: Concept-aligned Neurons for Visual Comparison of Neural Networks</b><br>S. Jeong et al., EuroVis                                                                               | 2024 |
| <b>Concept Lens: Visually Analyzing the Consistency of Semantic Manipulation in GANs</b><br>S. Jeong et al., IEEE VIS                                                                  | 2023 |
| <b>Interactively Assessing Disentanglement in GANs</b><br>S. Jeong et al., EuroVis                                                                                                     | 2022 |
| <b>Enhancing Work Trade Image Classification Performance Using a Work Dependency Graph</b><br>S. Jeong et al., Korean Journal of Construction Engineering and Management               | 2021 |
| <b>Comparing String Similarity Algorithms for Recognizing Task Names Found in Construction Documents</b><br>S. Jeong et al., Korean Journal of Construction Engineering and Management | 2020 |

## RESEARCH EXPERIENCE

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| <b>PhD Researcher</b> , Vanderbilt University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | January 2021 – Present |
| <ul style="list-style-type: none"><li>– Introduced a Bayesian framework that forecasts an LLM’s output distribution over structured Vega-Lite specifications from pre-generation hidden states, quantifying how much sampling variability is recoverable without repeated generation; identified channel binding as the main bottleneck across model scales (under review at ICLR 2027).</li><li>– Built an end-to-end evaluation framework for LLM-generated Vega-Lite visualizations (prompt construction, structured-output parsing, schema validation, rendering, feature-atom extraction); used it to quantify</li></ul> |                        |

how raw data, metadata, summaries, and stated goals shape design choices across thousands of generations.

- Developed concept-based interpretability methods for identifying, comparing, and steering semantic directions in GANs and diffusion models via representation analysis and intervention-based evaluation; methods released as visual analytics systems (Concept Lens, CAN) at IEEE VIS, TVCG, and EuroVis.
- Designed a text-based transfer function workflow that maps natural-language descriptions to semantic regions in volume rendering, bridging LLM embeddings with downstream generative pipelines (IEEE VIS 2024).
- Investigated LLM uncertainty calibration and persona-conditioned sampling to characterize how prompt framing shifts output distributions.

**Research Intern,** Lawrence Livermore National Laboratory

January 2023 – August 2023

- Built a scalable visual analytics system for exploring 80,000+ diffusion-generated candidate material structures, enabling domain scientists to triage and prioritize outputs that would otherwise require manual review.
- Partnered with materials science researchers to design the generative-model-driven discovery workflow end-to-end, from sampling to expert-in-the-loop evaluation.

**Senior Researcher,** Korea Institute of Industry Convergence

October 2020 – January 2021

- Applied NLP and computer-vision methods to construction-domain documents and imagery for automated task-name recognition, progress tracking, and work-trade classification.

**Student Researcher,** Tonglab, Vanderbilt University

February 2019 – February 2020

- Developed a Gabor-filter-based CNN inspired by early visual cortex processing, improving noisy-image classification accuracy by 8–12% over standard baselines.

## SOFTWARE AND LEADERSHIP EXPERIENCE

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**Co-founder,** MovTrack (Mobile Startup)

October 2018 – January 2020

- Led backend, Android, and AWS deployment for an early-stage consumer mobile product; owned infrastructure across frontend, backend, and cloud.

**Software Engineer,** KICM

March 2018 – July 2018

- Built data-processing automation for construction management research and cost-estimation workflows.

**President,** Korean Student and Scholars Association at Vanderbilt

August 2021 – July 2024

- Led a 60-member graduate organization and organized 10+ academic, cultural, and community events.